

Solution Architecture

Date	29 June 2026
Team ID	N/A
Project Name	OptiCrop Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

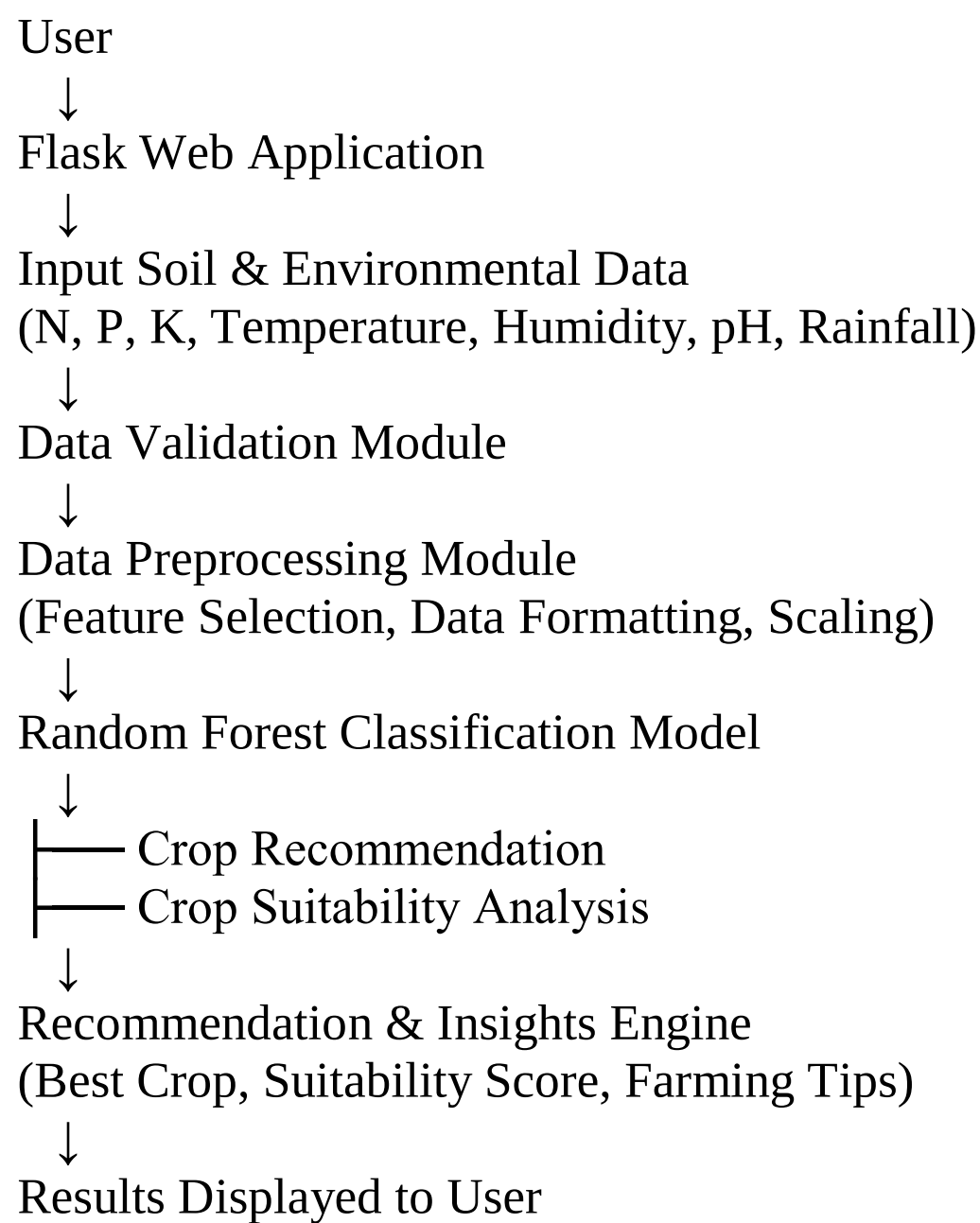
Solution ArchitectureDiagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Solution Architecture – OptiCrop

Architecture Flow



EXPLANATION:

User

The farmer, agricultural officer, researcher, or agribusiness user accesses the OptiCrop web application and provides soil and environmental information.

Flask Web Application

Acts as the interface between the user and the machine learning model, collecting inputs and displaying prediction results.

Input Soil & Environmental Data

The user enters Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall values.

Data Validation Module

Checks whether all input values are complete, valid, and within acceptable agricultural ranges before processing.

Data Preprocessing Module

Prepares the input data through formatting, feature selection, and scaling (if required) so it matches the model training format.

Random Forest Classification Model

Analyzes the processed data using a trained Random Forest classifier to identify the most suitable crop.

Crop Recommendation & Suitability Analysis

Generates the best crop recommendation and evaluates how suitable the current soil and climate conditions are for that crop.

Recommendation & Insights Engine

Provides the recommended crop, suitability level, and practical farming suggestions to improve productivity.

Results Displayed to User

Displays the final prediction and recommendations in a simple and user-friendly web interface.